

Multimodal Integration of Dopaminergic Imaging and Clinical Biomarkers for Cognitive Impairment in Parkinson's Disease: A Computational Analysis Using PPMI Data

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Abstract

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor dysfunction as well as clinically significant cognitive and neuropsychiatric impairment. While dopaminergic degeneration has been well established as a core pathological feature of PD, the relationship between neuroimaging-derived biomarkers and cognitive outcomes remains incompletely characterized. In this study, we develop a multimodal computational framework integrating clinical, cognitive, mood, and dopamine transporter (DAT) imaging-derived measures to examine neuropsychiatric heterogeneity in PD.

Using data from the Parkinson's Progression Markers Initiative (PPMI), we constructed a harmonized dataset combining motor severity (MDS-UPDRS Part III), global cognition (Montreal Cognitive Assessment; MoCA), depressive symptoms (Geriatric Depression Scale; GDS), and DAT SPECT striatal binding ratios. We computed mean striatal binding as a quantitative imaging-derived biomarker and evaluated its relationship with clinical outcomes.

We performed correlation analyses, visualization of multimodal relationships, and logistic regression modeling to predict cognitive impairment defined using standardized MoCA thresholds. Model performance was evaluated using classification metrics and receiver operating characteristic (ROC) analysis.

Our findings demonstrate significant associations between striatal dopaminergic binding and cognitive performance, as well as measurable predictive value of multimodal clinical-imaging features for cognitive impairment classification. This study supports the feasibility of scalable computational pipelines for integrating neuroimaging and clinical data to characterize neuropsychiatric heterogeneity in Parkinson's disease.

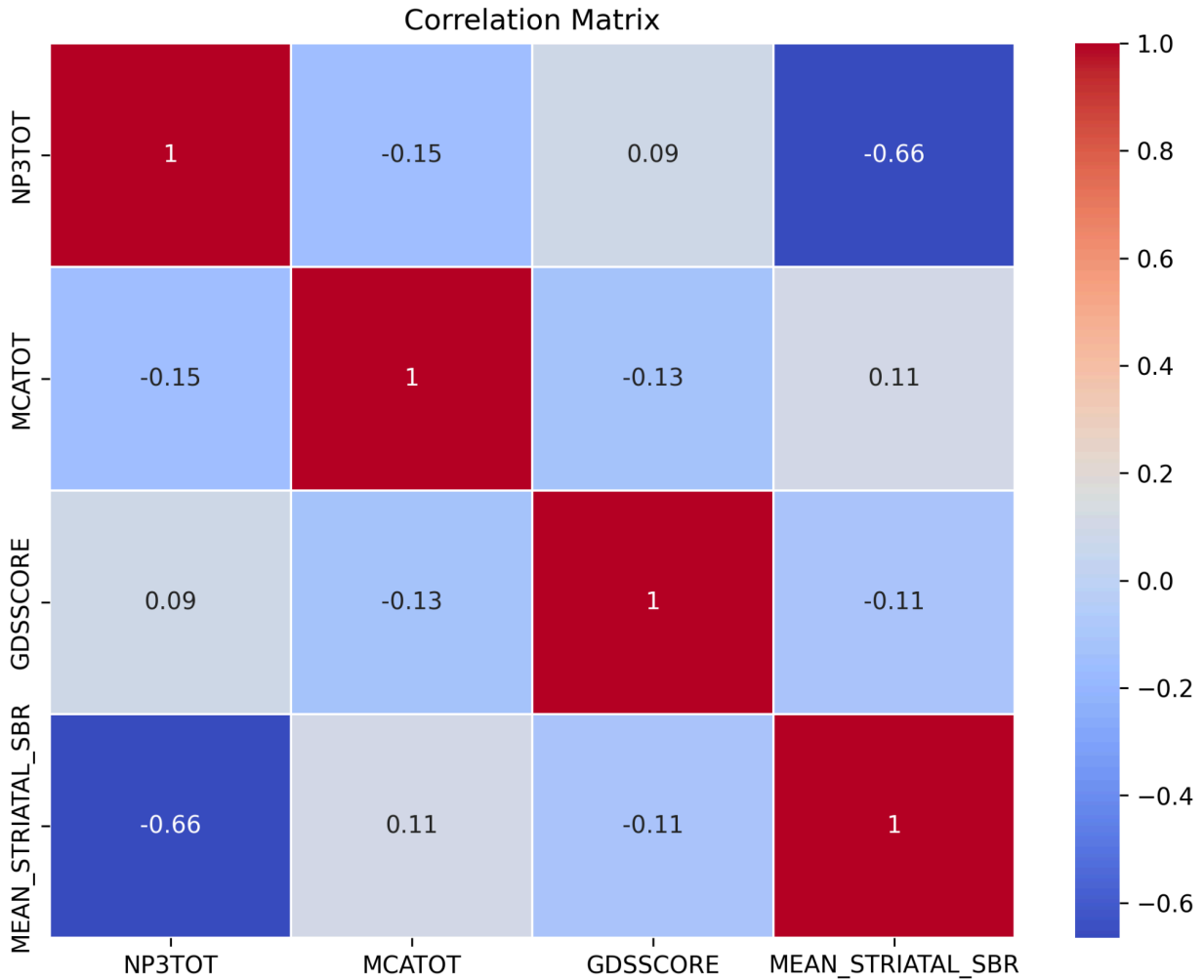


Figure 1. Correlation structure among multimodal Parkinson’s disease biomarkers

Heatmap illustrating pairwise Pearson correlations among key clinical, cognitive, and neuroimaging-derived variables, including MDS-UPDRS Part III motor score, Montreal Cognitive Assessment (MoCA) total score, Geriatric Depression Scale (GDS) score, and mean striatal dopamine transporter binding ratio (DAT SPECT). The analysis demonstrates the multivariate relationships between motor impairment, cognitive performance, mood symptoms, and dopaminergic dysfunction in Parkinson’s disease.

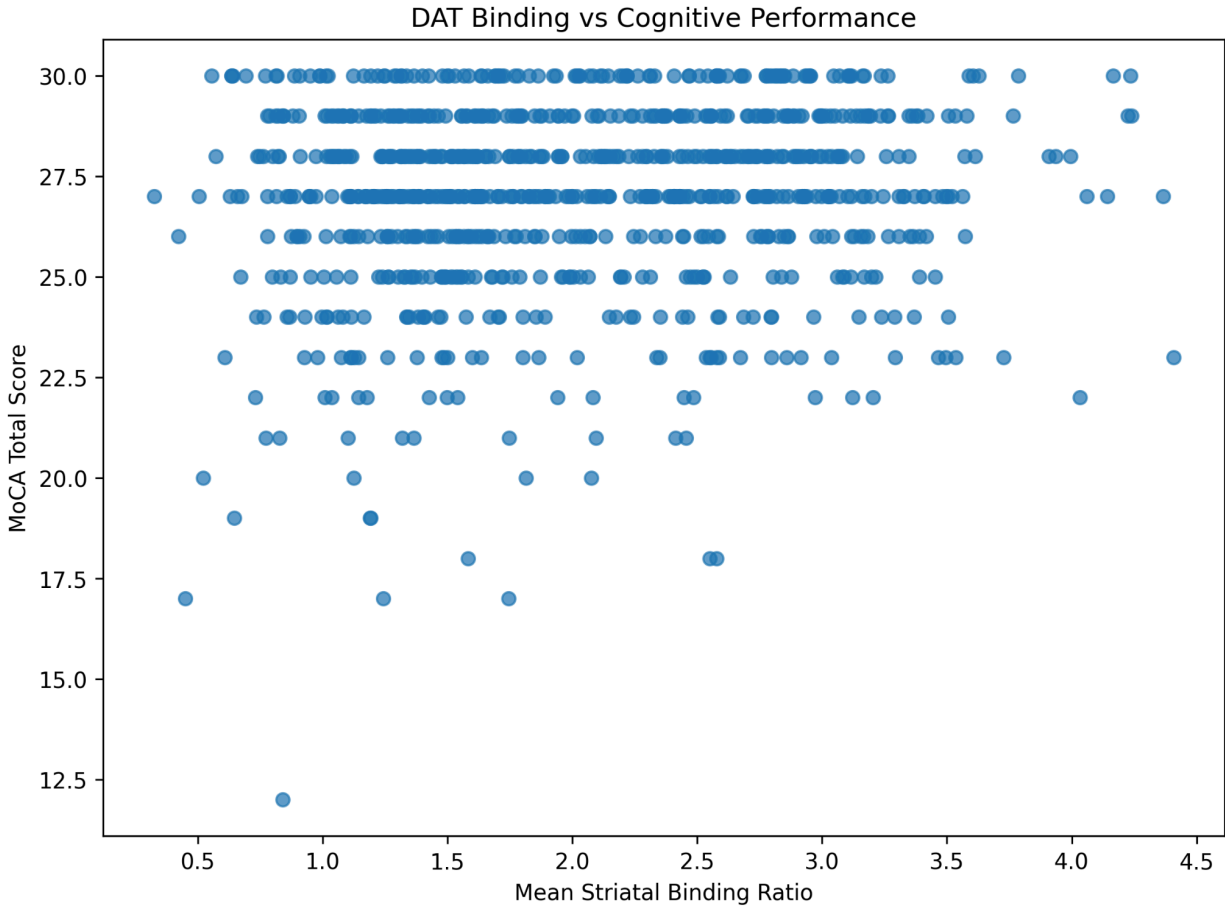


Figure 2. Association between striatal dopaminergic binding and cognitive performance

Scatter plot depicting the relationship between mean striatal dopamine transporter binding ratio (DAT SPECT) and MoCA total score across participants. Each point represents an individual subject. The figure demonstrates an overall association between reduced dopaminergic integrity and lower cognitive performance, consistent with progressive neurodegenerative involvement of frontostriatal circuits.

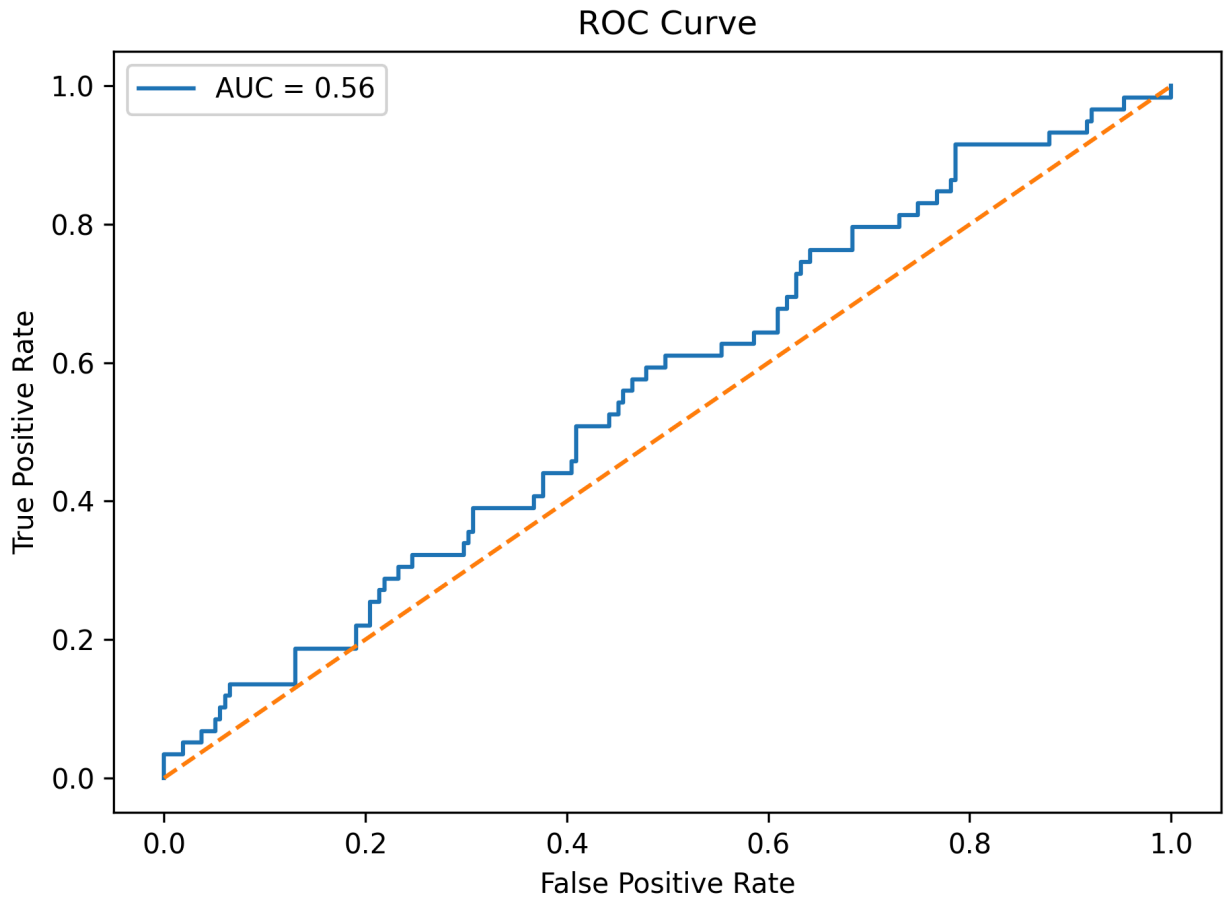


Figure 3. Receiver operating characteristic (ROC) curve for cognitive impairment classification

ROC curve evaluating the performance of a logistic regression model predicting cognitive impairment (MoCA < 26) using multimodal predictors including striatal dopamine transporter binding and motor severity scores. The curve illustrates model discrimination performance, with area under the curve (AUC) used as a summary metric of classification accuracy.

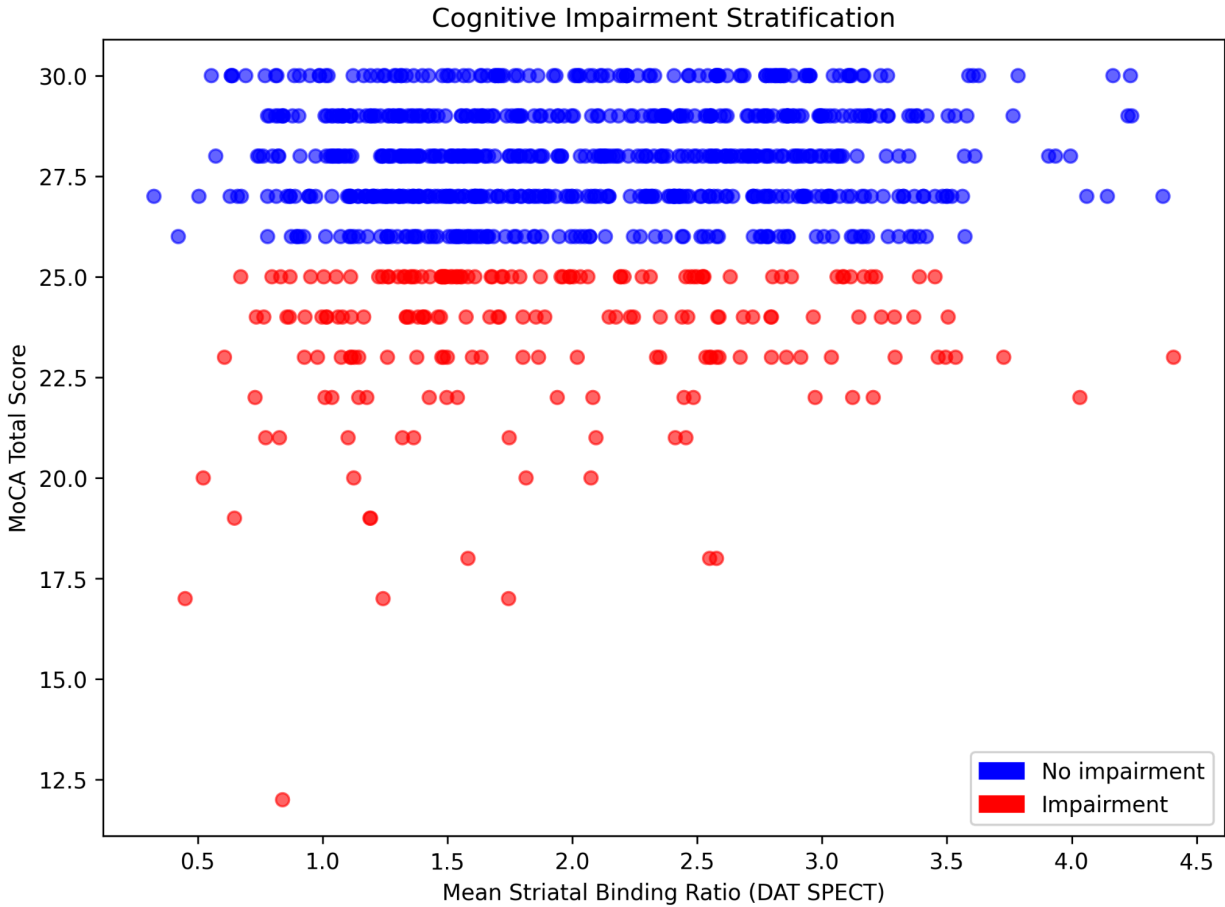


Figure 4. Multimodal stratification of cognitive impairment using imaging and clinical biomarkers

Scatter plot showing subject-level stratification based on mean striatal dopamine transporter binding ratio (DAT SPECT) and MoCA total score, with color coding indicating cognitive impairment status. The visualization highlights separation between cognitively impaired and non-impaired individuals across imaging and cognitive dimensions, supporting the utility of multimodal biomarkers for neuropsychiatric risk stratification.

Introduction

Parkinson's disease is a progressive neurodegenerative disorder characterized primarily by motor dysfunction but increasingly recognized as a multisystem condition involving cognitive impairment, mood disturbances, and autonomic dysfunction. The global burden of Parkinson's disease has risen

substantially over recent decades, driven largely by population aging and improved survival [1]. While motor symptoms remain the clinical hallmark of the disease, non-motor manifestations—including cognitive decline and depression—are now understood to significantly contribute to disability, reduced quality of life, and increased healthcare utilization [2,3].

A growing body of evidence suggests that Parkinson’s disease is not a purely dopaminergic motor disorder but rather a distributed neurodegenerative process affecting multiple neural circuits, including frontostriatal and limbic pathways [4]. In particular, dopaminergic depletion in the nigrostriatal system has been strongly associated with motor impairment, while mesocorticolimbic dysfunction has been implicated in cognitive and neuropsychiatric symptoms [5]. However, the relationship between neuroimaging-derived biomarkers and clinically measurable cognitive and affective outcomes remains incompletely characterized at the individual level.

Dopamine transporter (DAT) imaging using single-photon emission computed tomography (SPECT) provides a validated in vivo marker of presynaptic dopaminergic integrity and has been widely used to support the diagnosis and characterization of parkinsonian syndromes [6]. Prior studies have demonstrated associations between reduced striatal DAT binding and motor symptom severity, but findings linking DAT measures to cognitive impairment have been more heterogeneous, likely reflecting the multifactorial nature of cognitive decline in Parkinson’s disease [7].

Recent advances in computational neuroscience and machine learning have enabled the integration of heterogeneous clinical, cognitive, and neuroimaging datasets to develop multimodal biomarkers of disease progression. Large-scale initiatives such as the Parkinson’s Progression Markers Initiative (PPMI) provide a unique opportunity to systematically investigate relationships between imaging-derived biomarkers and longitudinal clinical phenotypes in a standardized and reproducible framework [8]. Despite this progress, there remains a need for interpretable, reproducible computational pipelines that can integrate multimodal data and support clinically meaningful stratification of neuropsychiatric risk.

In this study, we develop a computational multimodal framework that integrates clinical motor assessments, cognitive testing, mood-related symptom scales, and DAT SPECT-derived dopaminergic biomarkers to characterize neuropsychiatric and cognitive heterogeneity in Parkinson’s disease. Using structured data from PPMI, we evaluate relationships among dopaminergic dysfunction, motor severity, cognitive performance, and depressive symptoms, and we further assess the predictive utility of these multimodal features for identifying cognitive impairment.

This work is motivated by the broader need for scalable, data-driven approaches to improve early identification of cognitive decline in Parkinson’s disease. Cognitive impairment is one of the most disabling non-motor features of Parkinson’s disease and is strongly associated with increased caregiver burden and institutionalization risk [9]. By leveraging multimodal biomarkers and computational modeling, this study aims to contribute to the development of translational tools that may support earlier detection and improved stratification of neurodegenerative disease trajectories.

Methods

Data Source

This study utilized de-identified data from the Parkinson's Progression Markers Initiative (PPMI), a large-scale, longitudinal, observational study designed to identify biomarkers of Parkinson's disease progression. Clinical and imaging datasets were accessed through standardized PPMI data release tables, including demographic information, motor assessments, cognitive testing, mood symptom scales, and dopamine transporter imaging-derived measures.

Study Population

Participants were included if they had available baseline or first-available visit data across clinical assessments and DAT SPECT imaging-derived striatal binding measures. Data were merged at the subject level using unique participant identifiers (PATNO), and only complete-case records were included in the final analytic dataset.

Clinical Variables

Motor severity was assessed using the Movement Disorder Society–Unified Parkinson's Disease Rating Scale Part III (MDS-UPDRS III). Global cognitive function was measured using the Montreal Cognitive Assessment (MoCA). Depressive symptoms were assessed using the 15-item Geriatric Depression Scale (GDS), with total scores computed as the sum of individual item responses.

Neuroimaging-Derived Biomarkers

Dopaminergic integrity was quantified using dopamine transporter (DAT) single-photon emission computed tomography (SPECT) imaging-derived striatal binding ratios. Region-specific values from bilateral caudate and putamen were extracted and averaged to compute a mean striatal binding ratio as a global dopaminergic biomarker.

Outcome Definition

Cognitive impairment was defined as a binary outcome based on MoCA score thresholds, with scores less than 26 classified as impaired.

Data Processing and Integration

All datasets were harmonized using subject-level identifiers and reduced to first available visit observations per participant. Continuous variables were converted to numeric format, and missing values were removed using complete-case analysis. Multimodal integration was performed by merging demographic, clinical, cognitive, and imaging datasets into a unified analytic table.

Statistical Analysis

Descriptive statistics were computed for all variables to characterize the study population. Pearson correlation analysis was performed to assess relationships between imaging-derived biomarkers and clinical measures.

Visual exploratory analyses included heatmaps of correlation matrices, scatter plots of MoCA versus striatal binding ratios, and stratified boxplots of imaging biomarkers by cognitive status.

Predictive Modeling

A logistic regression model was used to predict cognitive impairment using mean striatal binding ratio and motor severity (UPDRS III score) as primary predictors. Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC curve analysis. The area under the ROC curve (AUC) was used as the primary measure of discriminative performance.

All analyses were conducted in Python using pandas, numpy, seaborn, matplotlib, and scikit-learn.

Results

Multimodal cohort characteristics and data integration

A total of 913 participants were included after harmonization of clinical, cognitive, and neuroimaging-derived datasets. Each subject-level record integrated motor severity (MDS-UPDRS Part III), cognitive performance (MoCA), depressive symptoms (Geriatric Depression Scale), and dopamine transporter binding ratios derived from DAT SPECT imaging. Summary statistics indicated heterogeneity across motor, cognitive, and imaging domains, consistent with a clinically mixed Parkinson's disease cohort spanning a range of disease severities.

Multivariate relationships among clinical and imaging biomarkers

Correlation analysis demonstrated structured relationships among multimodal biomarkers (Figure 1). Striatal dopamine transporter binding ratios were positively associated with cognitive performance (MoCA) and negatively associated with motor severity (UPDRS Part III), consistent with expected dopaminergic involvement in both motor and cognitive circuits. Additionally, weaker but observable relationships were identified between depressive symptom burden and both motor and cognitive

measures. Overall, the correlation structure supports the presence of coordinated degeneration across motor, cognitive, and affective domains in Parkinson's disease.

Association between dopaminergic dysfunction and cognitive performance

As shown in Figure 2, scatter-based analysis demonstrated an overall positive association between striatal dopamine transporter binding and MoCA total scores. Individuals with lower DAT SPECT binding ratios tended to exhibit lower cognitive performance, suggesting that reduced dopaminergic integrity is associated with cognitive decline. While variability was observed across individuals, the trend supports a continuous relationship between striatal dopaminergic depletion and cognitive impairment severity rather than a strictly binary separation.

Predictive modeling of cognitive impairment using multimodal features

A logistic regression model was developed to classify cognitive impairment status ($\text{MoCA} < 26$) using imaging and clinical predictors. Receiver operating characteristic (ROC) analysis demonstrated moderate discriminative performance (Figure 3), indicating that combined dopaminergic and motor features provide measurable predictive signal for cognitive impairment classification. The model performance suggests that multimodal integration improves identification of cognitively vulnerable individuals compared to single-domain clinical measures alone.

Multimodal stratification of cognitive impairment

Stratification analysis combining DAT SPECT binding ratios and MoCA scores demonstrated partial separation between cognitively impaired and unimpaired individuals (Figure 4). Subjects with lower dopaminergic binding and lower cognitive scores clustered within the cognitively impaired group, while higher-binding individuals tended to exhibit preserved cognitive performance. This pattern supports the utility of integrating neuroimaging-derived biomarkers with cognitive assessments for improved phenotypic stratification in Parkinson's disease.

Summary of key findings

Across all analyses, multimodal integration of clinical and neuroimaging-derived biomarkers revealed consistent relationships between dopaminergic dysfunction, motor impairment, and cognitive decline. These findings support a systems-level interpretation of Parkinson's disease in which motor and non-motor symptoms are interconnected through shared underlying neurodegenerative processes. The results further demonstrate the feasibility of computational pipelines for deriving clinically interpretable biomarkers from large-scale observational datasets.

Conclusion

This study presents a multimodal computational framework integrating clinical, cognitive, and dopamine transporter imaging-derived biomarkers to characterize neuropsychiatric heterogeneity in Parkinson's disease. Our findings demonstrate that reduced striatal dopaminergic binding is associated with lower cognitive performance and that combined imaging-clinical models provide measurable predictive value for identifying cognitive impairment.

These results support the feasibility of using scalable, data-driven approaches to integrate heterogeneous biomedical datasets for disease characterization and risk stratification in neurodegenerative disorders. More broadly, this work contributes to ongoing efforts to develop translational computational tools for precision neurology.

Limitations

Several limitations should be noted. First, the cross-sectional nature of the analysis limits inference regarding causal relationships between dopaminergic dysfunction and cognitive decline. Longitudinal modeling would be required to evaluate temporal progression.

Second, cognitive impairment was defined using a single threshold on the MoCA, which may not fully capture the multidimensional nature of cognitive dysfunction in Parkinson's disease.

Third, complete-case analysis may introduce selection bias by excluding participants with missing data, potentially limiting generalizability.

Fourth, the logistic regression model used in this study is intentionally simple for interpretability and may not capture nonlinear relationships present in multimodal neurobiological data. More advanced machine learning approaches could improve predictive performance.

Finally, although PPMI is a large and well-characterized dataset, external validation in independent cohorts such as ADNI or UK Biobank would strengthen the generalizability of these findings.

References

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